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Department of Physics

What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 4: Initial States and Free Modes

Group-B

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1 Free modes and canonical initial data

At zero mixing the Bunch–Davies light mode and momentum are

$$Q_{\mathrm{BD}} = \frac{(1-ix)e^{ix}}{\sqrt{2}}, \qquad \Pi_{q,\mathrm{BD}} = -\frac{e^{ix}}{\sqrt{2}x}. \tag{1}$$

For the spectator,

$$\Sigma_{\mathrm{BD}} = \frac{\sqrt{\pi}}{2} e^{i\pi(\nu/2+1/4)} x^{3/2} H_\nu^{(1)}(x), \quad \nu = \sqrt{9/4 - M^2}, \quad \Pi_{\sigma,\mathrm{BD}} = -\Sigma'_{\mathrm{BD}}/x^2. \tag{2}$$

For $M > 3/2$, ν is imaginary. We evaluate the Hankel functions with complex-order support and use $H'_\nu = (H_{\nu-1} - H_{\nu+1})/2$. The exponential prefactor supplies the required normalisation. At a finite initial X, the product initial state has canonical columns $(Q, \Pi_q, 0, 0)^T$ and $(0, 0, \Sigma, \Pi_\sigma)^T$. In the velocity form, $\dot{Q} = x^3\Pi_q - g\Sigma$, so the second column starts with $\dot{Q} = -g\Sigma$. The free analytic/numerical comparison on $10^{-3} \le x \le 100$ gives maximum scaled complex errors 5.805×10^{-10} (light) and 5.860×10^{-10} (heavy). The late light power is 0.5000005, consistent with $(1+x^2)/2 \to 1/2$.

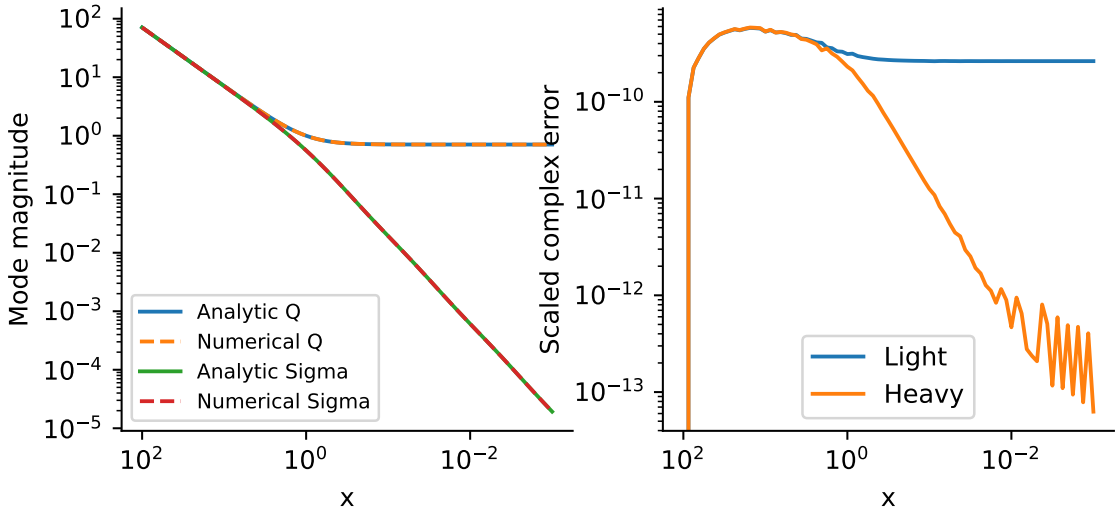


Figure 1: Free analytic and numerical mode magnitudes, together with errors in the complex modes.

2 Starting-time and commutator checks

With rows $(Q, \Pi_q, \Sigma, \Pi_\sigma)$, form $W = \mathcal{M}\mathcal{M}^\dagger$ and test $2\,\mathrm{Im}\,W = \Omega$, where $\Omega = \mathrm{diag}(J, J)$ and $J_{12} = 1$, $J_{21} = -1$. We require a maximum entrywise residual below 10^{-6} on the interval used for entropy. The free test reaches 5.225×10^{-7}. No final covariance renormalisation is applied.

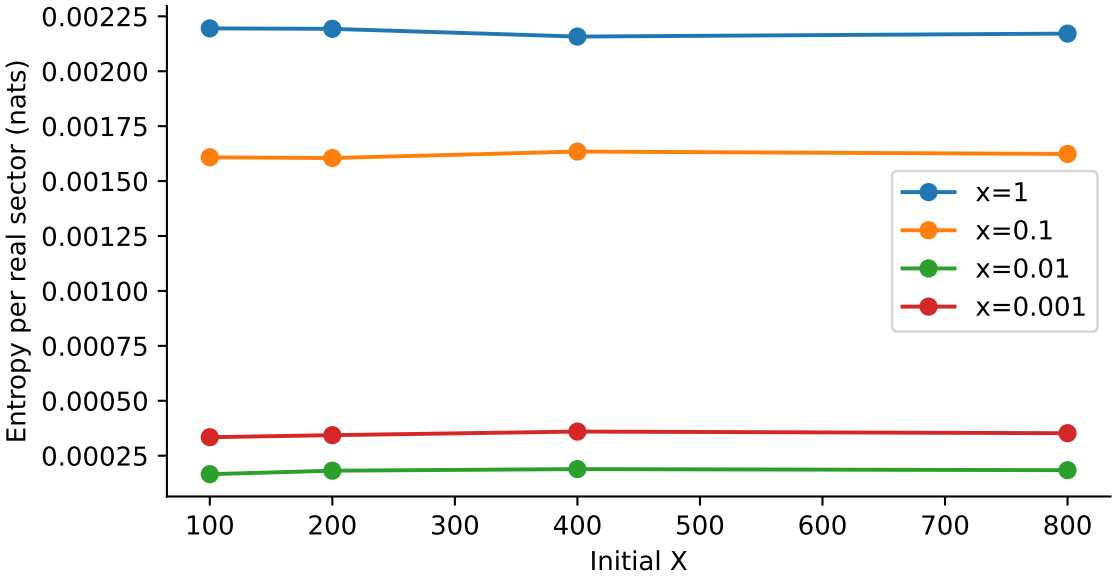


Figure 2: Finite-time product-state entropy under increasing initial X. These are diagnostic runs; late samples that fail the commutator criterion are not used as production entropy results.

The product-state limit is slow enough to matter for the much smaller bispectrum. We therefore also test a positive-frequency ultraviolet initial state, obtained from an asymptotic Riccati expansion. Its position covariance and momentum relation enforce the canonical initial commutators; it is not a correction imposed on the evolved state. Step 8 derives this construction. The saved adiabatic table compares orders 4 and 6 at $X = 100, 200, 400, 800$, including both power and entropy. At $x = 0.01$, the sixth-order entropy is approximately 1.845987×10^{-4} across these starting times. The production Gaussian scan uses order 6, $X = 400$ and tighter tolerances $2.5 \times 10^{-14}/10^{-17}$.

References

[1] X. Chen and Y. Wang, *Quasi-Single Field Inflation and Non-Gaussianities*, JCAP 04 (2010) 027, arXiv:0911.3380.

Code and supporting data: step4 on GitHub.

Contributions (equal)

Member	Contribution
Priyansh Rawat	Bunch–Davies modes and initial-state conventions.
Samarth Bhatnagar	Canonical commutator and starting-time checks.
Mayank Sinha	Free-mode implementation, adiabatic tests and figures.